VISVESVARAYA TECHNOLOGICAL UNIVERSITY
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Project Report
on

SYSTEM DESIGN FOR DIAGNOSTIC DEVICE SPIROMETER

Submitted in partial fulfillment of the requirements for the III semester
Mini Project Work [EC221]
Bachelor of Engineering
in
Electronics and Communication Engineering
of
Visvesvaraya Technological University, Belagavi.
by

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CERTIFICATE

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We also declare that, to the best of our knowledge and belief, the work reported here does not form part of any other report on the basis of which a degree or award was conferred on an earlier occasion on this by any other student.

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ABSTRACT

The Smart Spirometer is a health monitoring device designed to measure lung capacities and respiratory health. The system utilizes an STM32 microcontroller, a BMP280 pressure sensor, and three IR sensors to assess and monitor lung capacities at three specific thresholds: 600 ml, 900 ml, and 1200 ml. The BMP280 pressure sensor measures airflow pressure, while the IR sensors detect the corresponding lung volume during exhalation. An LCD screen provides real-time feedback to the user, displaying their lung capacity and indicating the achieved thresholds. This project aims to assist in the early detection of respiratory issues, facilitate regular health monitoring, and provide feedback on lung performance for patients with conditions like asthma, chronic obstructive pulmonary disease (COPD), or post-recovery from respiratory illnesses. The compact design, precise measurements, and ease of use make the Smart Spirometer an essential tool for healthcare monitoring in clinical and personal environments.

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CHAPTER 1

INTRODUCTION

GOAL:

Lung capacity and respiratory health are critical indicators of overall well-being, especially for individuals recovering from respiratory illnesses, managing chronic conditions such as asthma or chronic obstructive pulmonary disease (COPD), or enhancing their athletic performance. Traditional spirometers, while effective, are often expensive, bulky, or limited in accessibility. This has created a need for a cost-effective, portable, and user-friendly device that can reliably monitor lung capacity. The Smart Spirometer addresses this need by integrating modern sensors and microcontroller technology to deliver accurate and real-time measurements of lung capacity. The system is built around the STM32 microcontroller, renowned for its high processing power and energy efficiency. A BMP280 pressure sensor is used to measure the pressure changes caused by airflow during exhalation. Additionally, three IR sensors are strategically calibrated to detect lung capacity thresholds of 600 ml, 900 ml, and 1200 ml, allowing users to track their respiratory performance in a structured manner. The device features an LCD to display real-time results, offering immediate feedback to users. This portable and easy-to-operate spirometer is designed to support regular monitoring in home settings, providing individuals and healthcare providers with essential data for respiratory health management. Its compact design and integration of low-cost components ensure accessibility for a wide range of users. By enabling users to monitor their lung capacity conveniently, the Smart Spirometer aims to promote proactive health management, facilitate early detection of respiratory issues, and support rehabilitation efforts for patients with compromised lung function.

OBJECTIVES:

- **Develop a Portable Lung Monitoring Device**

  Design a compact and user-friendly spirometer using the STM32 microcontroller to enable accurate measurement of lung capacity.
- **Incorporate Accurate Lung Capacity Detection**

  Utilize a BMP280 pressure sensor to measure airflow pressure and IR sensors to detect predefined lung capacity thresholds of 600 ml, 900 ml, and 1200 ml.
- **Provide Real-Time Feedback**

  Display real-time lung capacity readings and threshold achievements on an LCD for immediate user feedback.
- **Enhance Accessibility and Affordability**

  Create a cost-effective solution suitable for both clinical and personal use, promoting accessibility for individuals across different socioeconomic backgrounds.
- **Support Early Detection and Monitoring**

  Enable regular monitoring to assist in the early detection of respiratory issues, such as asthma or chronic obstructive pulmonary disease (COPD), and track progress during recovery or rehabilitation.
- **Promote Proactive Health Management**

  Encourage users to monitor and manage their respiratory health through an easy-to-operate device that supports health awareness and preventive care.

Lung diseases such as asthma, chronic obstructive pulmonary disease (COPD), and other respiratory conditions are among the leading causes of morbidity and mortality worldwide. Early diagnosis and continuous monitoring of lung function are crucial for effective management and prevention of these diseases. Traditional spirometers, which measure lung volume and airflow, are valuable tools in assessing pulmonary function but often require clinical visits for accurate measurements. This can lead to delays in diagnosis, limited access to healthcare, and poor patient adherence to monitoring.

To address these challenges, the development of a smart spirometer system offers a promising solution. This project aims to create an advanced, portable, and user-friendly spirometer that integrates modern technologies such as sensor systems, real-time data processing, wireless connectivity, and cloud-based health monitoring to enable continuous, at-home lung function monitoring. By utilizing advanced sensors (e.g., ultrasonic flow sensors, pressure sensors), the system will ensure accurate measurements of lung function parameters like forced expiratory volume (FEV1) and forced vital capacity (FVC). These measurements will be processed through intelligent algorithms to provide users with real-time feedback on their performance and lung health.

Beyond basic spirometry, the smart system will be equipped with features like mobile app integration, allowing users to track their lung function over time, receive personalized health recommendations, and share their results securely with healthcare providers for remote monitoring. With AI-driven insights, the system can alert users to any significant changes in their lung function, promoting proactive care and timely medical intervention.

The project seeks to address several key challenges in lung disease management, including ensuring accurate measurements, providing personalized health insights, enhancing patient engagement, and facilitating remote monitoring. By leveraging Internet of Things (IoT) technology, data security protocols, and cloud platforms, the device aims to empower patients, improve healthcare accessibility, and support healthcare professionals in managing respiratory health more effectively. Ultimately, this smart spirometer system is designed to revolutionize how patients monitor and manage their lung health, offering a seamless, reliable, and accessible solution for early detection and chronic disease management.

RESEARCH METHODOLOGY:

1. Problem Identification: Address limitations of traditional spirometers like accessibility and accuracy.
2. Literature Review: Study existing technologies, respiratory care needs, and global standards.
3. Design and Development: Create hardware with sensors and software for data analysis, focusing on usability and real-time feedback.
4. Testing and Validation: Perform technical, clinical, and usability tests to ensure accuracy, reliability, and compliance with standards.
5. Data Collection and Analysis: Analyze performance data from trials and compare results with traditional devices.
6. Feedback and Refinement: Use patient and expert feedback to iteratively improve the device.
7. Deployment: Manufacture and distribute the device, ensuring cost-effectiveness and user training.
8. Post-Deployment Evaluation: Monitor real-world performance and gather insights for future enhancements.

CHAPTER 2

Literature Survey

- **"Development of a Portable Spirometer with MPX5500DP Air Pressure Sensor and Atmega328 Microcontroller"**
- Discusses the design of a low-cost, portable spirometer using pressure sensors and microcontrollers for respiratory health monitoring.
- **"Design and Development of Low-Cost Spirometer with PC Interface"**
- Explores creating an affordable spirometer that interfaces with a computer, utilizing pressure sensors for accurate measurements.
- **"Portable Spirometer for Measuring Lung Function Health (FVC and FEV1)"**
- Focuses on building an affordable device to measure Forced Vital Capacity (FVC) and Forced Expiratory Volume in one second (FEV1).
- **"Non-Invasive Spirometer Using STM32 and Bluetooth-MEMS Sensors"**
- Highlights the use of STM32 microcontrollers and MEMS sensors for creating a non-invasive spirometer.
- **"Design and Evaluation of a Novel Venturi-Based Spirometer for Accurate Respiratory Function Assessment"**
- Introduces a Venturi-based spirometer prototype to ensure precise respiratory measurements.
- **"Smart Spirometer for Post-COVID Rehabilitation"**
- Proposes an IoT-based spirometer for at-home rehabilitation and monitoring for post-COVID patients.
- **"Development of a Portable Ventilatory Function Measurement Device Using Embedded Sensors"**
- Discusses the integration of embedded sensors for lung function analysis in a portable device.
- **"Implementation of BMP280 Sensor for Accurate Spirometry in Wearable Devices"**
- Examines the use of BMP280 sensors for respiratory health monitoring in wearable applications.

- **"Low-Cost Embedded Spirometer Based on Commercial Micro-Machined Platinum Thin Film Sensors"**
- Focuses on creating an embedded spirometer for affordable healthcare applications.
- **"IoT-Based Smart Spirometer for Home Monitoring of Respiratory Health"**
- Presents an IoT-enabled spirometer for remote respiratory health assessments.

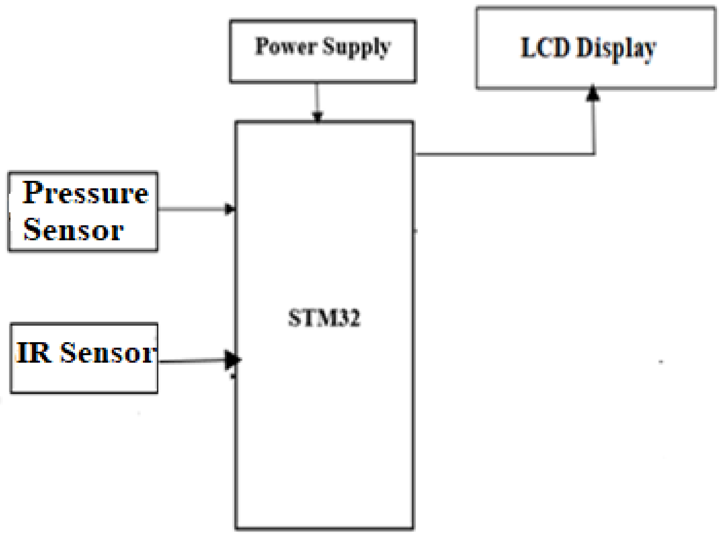
Problem Statement:

Respiratory health issues, such as asthma, chronic obstructive pulmonary disease (COPD), and other lung-related conditions, are prevalent and require regular monitoring to ensure effective management. Traditional spirometers, while effective, are often expensive, bulky, and inaccessible to individuals outside clinical settings. This lack of accessibility creates a gap in regular lung capacity monitoring, especially for individuals in remote or underserved areas. Moreover, existing devices often fail to provide real-time, user-friendly feedback, which limits their use in non-clinical environments. Patients recovering from respiratory illnesses or engaged in rehabilitation programs require affordable and portable solutions to track their progress effectively. The Smart Spirometer aims to address these challenges by providing a cost-effective, portable, and easy-to-use device capable of accurately measuring lung capacity at predefined thresholds. By leveraging modern microcontroller and sensor technology, the device seeks to bridge the gap between clinical-grade monitoring and personal healthcare needs, empowering users to take control of their respiratory health.

CHAPTER 3

3.1. METHODOLOGY

Block Diagram:



Working:

The Smart Spirometer is designed to measure lung capacity using an STM32 microcontroller, a BMP280 pressure sensor, and three IR sensors. Here's how the system works:

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1. Initialization and Calibration

- **Components Power Up**:
  - The STM32 initializes the BMP280 pressure sensor, IR sensors, and LCD.
  - Calibration is performed to ensure the BMP280 reads ambient air pressure accurately.
- **Sensor Configuration**:
  - The BMP280 is configured to measure pressure changes due to airflow during exhalation.

- IR sensors are calibrated to detect airflow levels corresponding to lung capacities (e.g., 600 ml, 900 ml, and 1200 ml).

---

2. User Interaction

- **Breathing Measurement**:
  - The user exhales into a tube connected to the spirometer.
  - The pressure from the exhaled air causes a measurable change in pressure detected by the BMP280 sensor.
- **IR Sensor Activation**:
  - The airflow passes through the tube, triggering the IR sensors sequentially.
  - Each IR sensor corresponds to a specific lung capacity (600 ml, 900 ml, 1200 ml).

---

3. Data Processing

- **Pressure Data**:
  - The BMP280 sensor measures pressure changes and sends the data to the STM32 via I2C or SPI communication.
  - The STM32 processes the pressure data to determine the strength and duration of exhalation.
- **IR Sensor Data**:
  - The STM32 detects the activation of IR sensors to estimate the lung capacity reached by the user.

---

4. Displaying Results

- The LCD displays:
  - **Lung Capacity**: Based on the highest IR sensor activated.
  - **Exhalation Pressure**: Calculated from the BMP280 data.
  - A **Message or Warning** if the lung capacity is below a threshold indicating a potential health issue.

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5. Additional Features

- **Data Logging**:
  - The system can store data on an SD card or transmit it via Bluetooth or Wi-Fi for long-term monitoring.
- **Alerts**:
  - If lung capacity is consistently below normal levels, the system can send an alert to a healthcare provider.
- **Energy Efficiency**:
  - The system is designed to be low power, with the STM32 operating in low-power modes when idle

CHAPTER 4

Hardware and Software Requirements

Hardware Requirements:

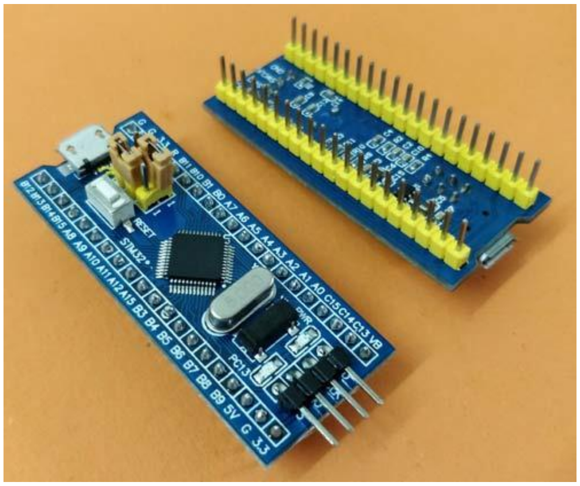
- **STM32**
- **IR Sensors**
- **Pressure Sensor**
- **Power Supply**
- **Spirometer**

Software Requirements:

- **Arduino IDE**
- **Embedded C**

STM32:

The STM32F103C8T6 (also known as 'STM32' or 'Blue Pill") is a cheap development board based on the ARM Cortex M3 microprocessor. This video by Great Scott can prove to be an introductory video to understand what it exactly is and how it can be used.



Naming Convention of STM microcontrollers

Parameter	Meaning
STM	name of the manufacturer (STMicroelectronics)
32	32 bit ARM architecture
F	Foundation
1	Core (ARM Cortex M3)
03	Line (describes peripherals and speed)
C	48 pins
8	64 KB flash memory
T	LQFP package (Low Profile Quad Flat Pack)
6	Operating Temperature Range (-40 °C to 85 °C)

Technical Specifications of STM32

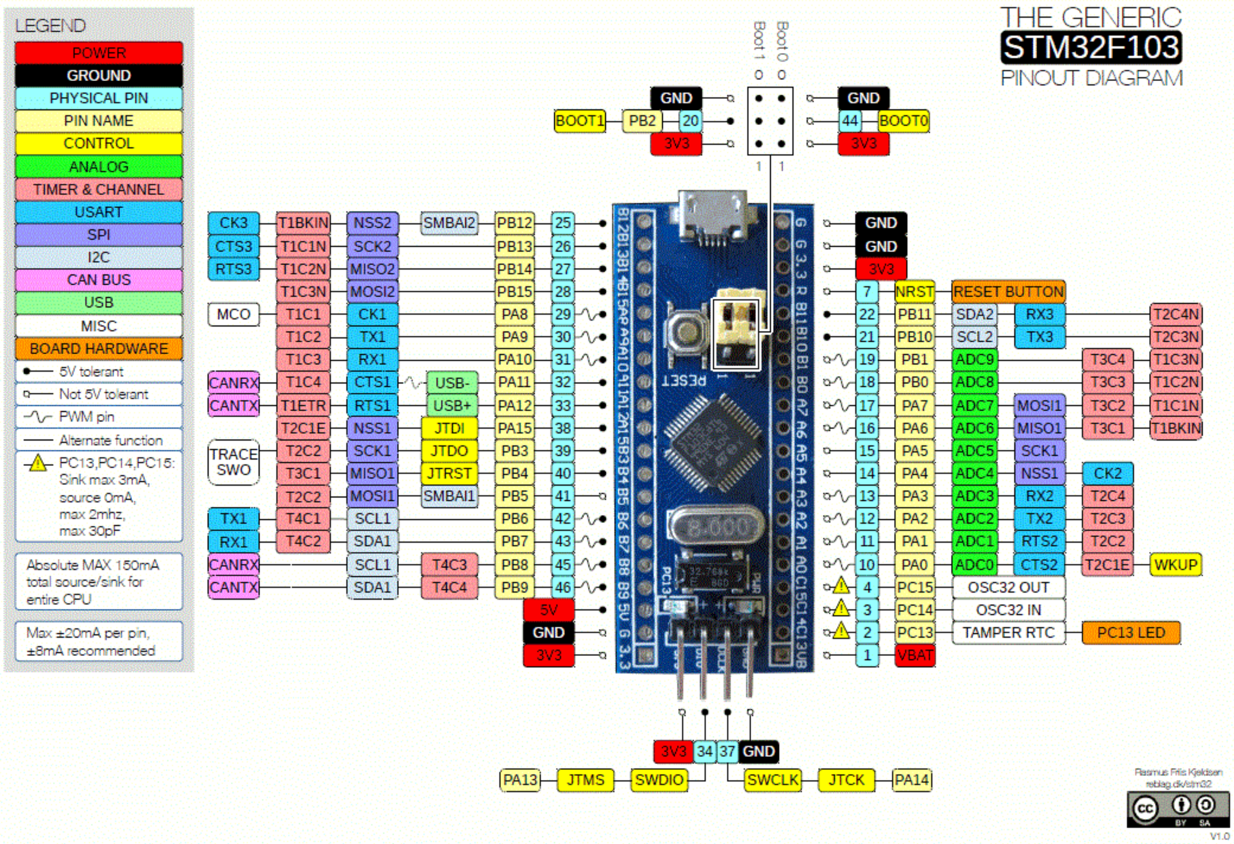
Parameter	Meaning
Architecture	32 bit ARM Cortex M3

Parameter	Meaning
Operating Voltage	2.7V to 3.6V
CPU Frequency	72 MHz
Number of GPIO pins	37
Number of PWM pins	12
Analog Input Pins	10 (12 bit resolution)
I2C Peripherals	2
SPI Peripherals	2
CAN 2.0 Peripheral	1
Timers	3(16-bit), 1
Flash Memory	64KB
RAM	20Kb

For more insights about the technical specifcations refer to the official datsheet and reference manual by

STMicroelectronics.

Pinout



Programming STM32

1) Using STM32duino bootloader (Arduino IDE)

You can program your STM32 development board using Arduino IDE, too. You will require FTDI (USB to UART converter) for this process. This tutorial explains the complete process.

2) Using Keil UVision and STM32CubeMX

This is one step further than the last mentioned process and is more professional in terms of usage. You will require the softwares ARM's Keil Uvision and STM32CubeMX for this method of programming BluePill. You will also need the STLink/V2 which is a debugger cum programmer hardware provided by STMicroelectronics. These softwares provide a more sophisticated and professional programming environment for programming embedded systems. You may refer to this guide to know this method in detail.

Understanding Blue Pill

The Blue Pill is a 32-bit Arduino compatible development board that features the STM32F103C8T6, a member of the STM32 family of ARM Cortex-M3 core microcontrollers. This board aims to bring the 32-bit ARM Cortex microcontrollers to the hobbyist market with the Arduino style form factor.

Powering your Blue Pill:

There are three ways of powering your Blue Pill development board:

- Using the built-in USB micro connector.
- Supplying 5V to the 5V pin as external supply.
- Supplying 3.3V directly to the 3.3V pin.

Input/Output:

The Blue Pill has 37 GPIO pins spread across four ports – A and B (16 pins), C (3 pins) and D (2 pins). Each pin has a current sink/source ability of 6mA. Pull-up and pull-down resistors can be enabled on each of the pins.

Most pins have extra functionality as well:

- Serial ports – receive and transmit data via the UART protocol
- I^2C ports – two-wire communication via the IIC protocol
- SPI – serial communication
- PWM
- Pin 13 has a built-in LED

These special functions and their respective pins are illustrated in the Blue Pill pin diagram shown above.

How to Use the STM32 Development Board?

The Blue Pill can be programmed in two ways:

- Using an external USB/Serial converter connected to UART1 pins, which is the default bootloader for this family of boards. It can be programmed using the Arduino software this way.
- STLink USB Dongle – this uses the single-wire debug interface to communicate with the board. This allows it to be programmed using advanced software like Keil/CubeMX. It also allows memory access using the STLink software.

Before programming, it is important to connect the BOOT0 jumper to 1 and press the reset button to put the chip in 'programming mode'.

Uploading Your First Program

If programming using the Arduino software, the appropriate board files should be downloaded using preferences and boards manager.

Then the correct board must be selected on the board's menu.

Since the built-in LED is on pin 13, just like the Arduino, the basic blink sketch will work on the Blue Pill.

Applications

- Rapid prototyping
- CRC calculations
- Robotics
- Consumer products
- Drone controllers

Infra red sensor

It is an electronic instrument. It is used to sense certain characteristics in its surroundings by emitting infrared radiations. Capable of measuring heat and detection of moving object.

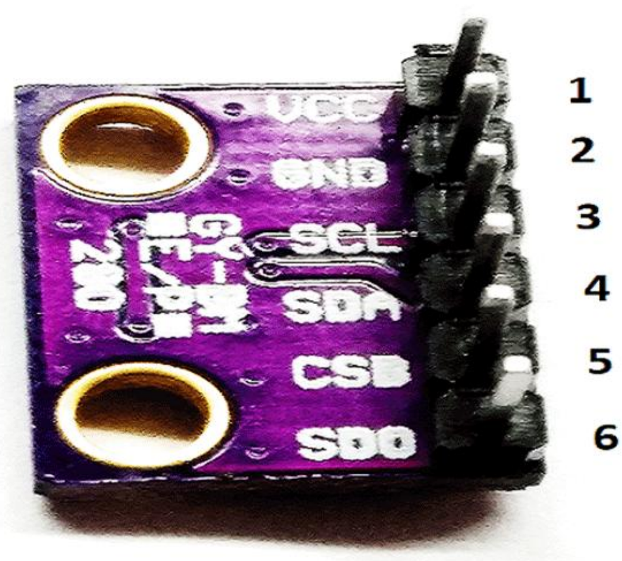
Fig 5.3 IR Sensor

This type of sensor measures only infrared radiation, rather than emitting it. Specifications: Size:3mm TX: Transmitter RX: Receive The living or non-living object can be detected using this sensor by sensing the heat from the object. It has 4 pins, vcc gnd and out. The operating voltage is 5V.

An IR sensor can measure the heat of an object as well as detects the motion. These types of sensors measures only infrared radiation, rather than emitting it that is called as a passive IR sensor. Usually in the infrared spectrum, all the objects radiate some form of thermal radiations.

BMP280

Pressure sensor



The BMP280 is an absolute barometric pressure sensor, which is especially feasible for mobile applications. Its small dimensions and its low power consumption allow for the implementation in battery-powered devices such as mobile phones, GPS modules or watches. The BMP280 is based on Bosch’s proven piezo-resistive

~~pressure sensor technology featuring high accuracy and linearity as well as long-term stability and~~

high EMC robustness. Numerous device operation options guarantee for highest flexibility. The device is optimized in terms of power consumption, resolution and filter performance.Robert Bosch is the world market leader for pressure sensors in automotive and consumer applications. Bosch's proprietary APSM (Advanced Porous Silicon Membrane) MEMS manufacturing process is fully CMOS compatible and allows a hermetic sealing of the cavity in an all silicon process. The BMP280 is based on Bosch's proven Piezo-resistive pressure sensor technology featuring high EMC robustness, high accuracy and linearity and long term stability. The BMP280 is an absolute barometric pressure sensor especially designed for mobile applications. The sensor module is housed in an extremely compact 8-pin metal-lid LGA package with a footprint of only 2.0 × 2.5 mm2 and 0.95 mm package height. Its small dimensions and its low power consumption of 2.7 µA @1Hz allow the implementation in battery driven devices such as mobile phones, GPS modules or watches. As the successor to the widely adopted BMP180, the BMP280 delivers high performance in all applications that require precise pressure measurement. The BMP280 operates at lower noise, supports new filter modes and an SPI interface within a footprint 63% smaller than the BMP180. The emerging applications of in-door navigation, health care as well as GPS refinement require a high relative accuracy and a low TCO at the same time. BMP180 and BMP280 are perfectly suitable for applications like floor detection since both sensors feature excellent relative accuracy is ±0.12 hPa, which is equivalent to ±1 m difference in altitude. The very low offset temperature coefficient (TCO) of 1.5 Pa/K translates to a temperature drift of only 12.6 cm/K. Please contact your regional Bosch Sensortec partner for more information about software packages enhancing the calculation of the altitude given by the BMP280 pressure reading.

Table 1: Comparison between BMP180 and BMP280

Parameter	BMP180	BMP280
Footprint	3.6 × 3.8 mm	2.0 × 2.5 mm
Minimum V_{DD}	1.80 V	1.71 V
Minimum V_{DDIO}	1.62 V	1.20 V
Current consumption @3 Pa RMS noise	12 µA	2.7 µA
RMS Noise	3 Pa	1.3 Pa
Pressure resolution	1 Pa	0.16 Pa
Temperature resolution	0.1°C	0.01°C
Interfaces	I²C	I²C & SPI (3 and 4 wire, mode '00' and '11')
Measurement modes	Only P or T, forced	P&T, forced or periodic
Measurement rate	up to 120 Hz	up to 157 Hz
Filter options	None	Five bandwidths

Block diagram

Figure 1 shows a simplified block diagram of the BMP280:

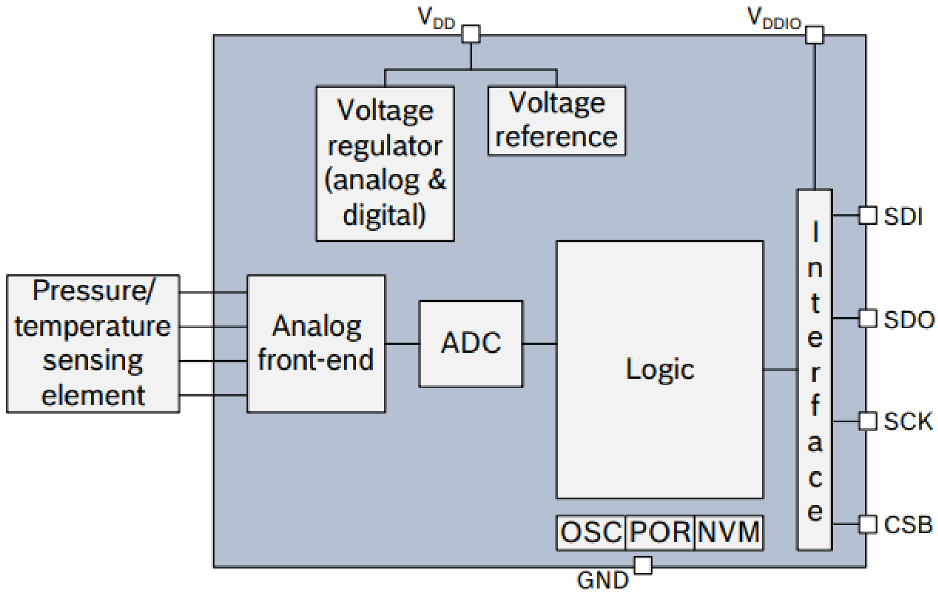


Figure 1: Block diagram of BMP280

The BMP280 has two separate power supply pins

• VDD is the main power supply for all internal analog and digital functional blocks

• VDDIO is a separate power supply pin, used for the supply of the digital interface A power-on reset generator is built in which resets the logic circuitry and the register values after the power-on sequence. There are no limitations on slope and sequence of raising the VDD and VDDIO levels. After powering up, the sensor settles in sleep mode (see 3.6.1). Warning. Holding any interface pin (SDI, SDO, SCK or CSB) at a logical high level when VDDIO is switched off can permanently damage the device due caused by excessive current flow through the ESD protection diodes. If VDDIO is supplied, but VDD is not, the interface pins are kept at a high-Z level. The bus can therefore already be used freely before the BMP280 VDD supply is established.

Measurement flow The BMP280 measurement period consists of a temperature and pressure measurement with selectable oversampling. After the measurement period, the data are passed through an optional IIR filter, which removes short-term fluctuations in pressure (e.g. caused by slamming a door). The flow is depicted in the diagram below.

LCD display:

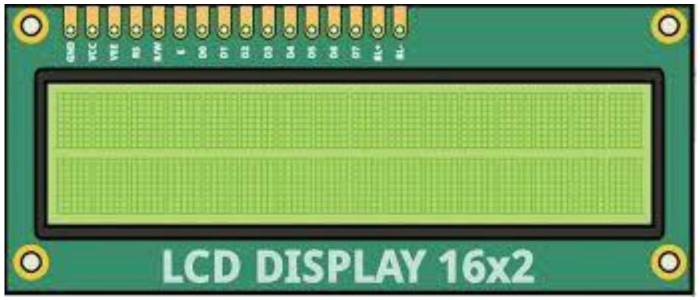


Fig . 16*2 LCD display

A **liquid-crystal display** (**LCD**) is a flat-panel display or other electronically modulated optical device that uses the light-modulating properties of liquid crystals. Liquid crystals do not emit light directly, instead using a backlight or reflector to produce images in color or monochrome. LCDs are available to display arbitrary images (as in a general-purpose computer display) or fixed images with low information content, which can be displayed or hidden, such as preset words, digits, and 7-segment displays, as in a digital clock. They use the same basic technology, except that arbitrary images are made up of a large number of small pixels, while other displays have larger elements.

LCD is used in wide range application including computer monitors, televisions, instrument panels, aircraft cockpit displays, and indoor and outdoor signage. Small LCD screens are common in portable consumer devices such as digital cameras, watches, calculators, and mobile telephones, including smartphones. LCD screens are also used on consumer electronics products such as DVD players, video game devices and clocks. LCD screens have replaced heavy, bulky cathode ray tube (CRT) displays in nearly all applications. LCD screens are available in a wider range of screen sizes than CRT and plasma displays, with LCD screens available in sizes ranging from tiny digital watches to huge, big-screen television sets.

Since LCD screens do not use phosphors, they do not suffer image burn-in when a static image is displayed on a screen for a long time (e.g., the table frame for an aircraft schedule on an indoor sign). LCDs are, however, susceptible to image persistence.

Interfacing an LCD with an Arduino:

The 16x2 LCD has a total of 16 pins. As shown in the table below, eight of the pins are data lines (pins 7-14), two are for power and ground (pins 1 and 16), three are used to control the operation of LCD (pins 4-6), and one is used to adjust the LCD screen brightness (pin 3). The remaining two pins (15 and 16) power the backlight.The details of the LCD terminals are as follows:

Terminal 1	GND
Terminal 2	+5V
Terminal 3	Mid terminal of potentiometer (for brightness control)
Terminal 4	Register Select (RS)
Terminal 5	Read/Write (RW)
Terminal 6	Enable (EN)
Terminal 7	DB0
Terminal 8	DB1
Terminal 9	DB2
Terminal 10	DB3
Terminal 11	DB4
Terminal 12	DB5
Terminal 13	DB6
Terminal 14	DB7
Terminal 15	+4.2-5V
Terminal 16	GND

LCD MODULE:

Fig . Output of LCD

The name and functions of each pin of the 16×2 LCD module is given below.

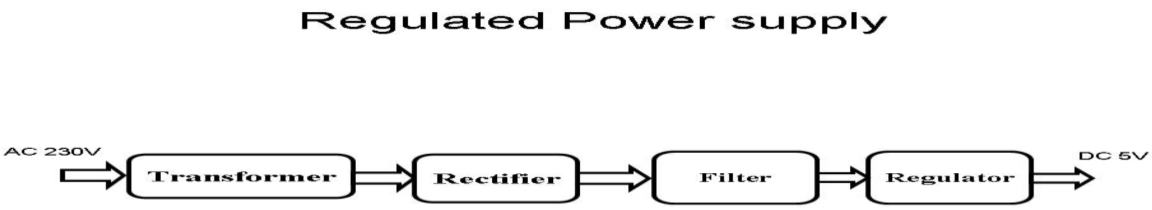
Pin1 (Vss): Ground pin of the LCD module.
Pin2 (Vcc): Power to LCD module (+5V supply is given to this pin)
Pin3 (VEE): Contrast adjustment pin. This is done by connecting the ends of a 10K potentimeter to +5V and ground and then connecting the slider pin to the VEE pin. The voltage at the VEE pin defines the contrast.The normal setting is between 0.4 and 0.9V.
Pin4(RS): Register select pin. Logic HIGH at RS pin selects data register and logic LOW at RS pin selects command register. If we make the RS pin HIGH and feed an input to the data lines (DB0 to DB7), this input will be treated as data to display on LCD screen. If we make the RS pin LOW and feed an input to the data lines, then this will be treated as a command (a command to be written to LCD controller – like positioning cursor or clear screen or scroll).
Pin5 (R/W): Read/Write modes. This pin is used for selecting between read and write modes. Logic HIGH at this pin activates read mode and logic LOW at this pin activates write mode.
Pin6 (E): This pin is meant for enabling the LCD module. A HIGH to LOW signal at this pin will enable the module.
Pin7 (DB0) to Pin14(DB7): These are data pins. The commands and data are fed to the LCD module though these pins.

Pin15 (LED+): Anode of the back light LED. When operated on 5V, a 560 ohm resistor should be connected in series to this pin. In Arduino based projects the back light LED can be powered from the 3.3V source on the Arduino board.

Pin16 (LED-): Cathode of the back light LED.

RS pin of the LCD module is connected to digital pin 12 of the Arduino. R/W pin of the LCD is grounded. Enable pin of the LCD module is connected to digital pin 11 of the Arduino. This method is very simple, requires less connections and we can almost utilize the full potential of the LCD module. Digital lines DB4, DB5, DB6 and DB7 are interfaced to digital pins 5, 4, 3 and 2 of the Arduino. The 10K potentiometer is used for adjusting the contrast of the display. The Arduino can be powered through the external power jack provided on the board. +5V required in some other parts of the circuit can be tapped from the 5V source on the Arduino board. The Arduino can be also powered from the PC through the USB port.

Regulated power supply:



Transformer:

A transformer is a device that transfers electrical energy from one circuit to another through inductively coupled conductors without changing its frequency. A varying current in the first or primary winding creates a varying magnetic flux in the transformer's core, and thus a varying magnetic field through the secondary winding. This varying magnetic field induces a varying electromotive force (EMF) or "voltage" in the secondary winding. This effect is called mutual induction. If a load is connected to the secondary, an electric current will flow in the secondary winding and electrical energy will be transferred from the primary circuit through the transformer to the load. This field is made up from lines of force and has the same shape as a bar magnet. If the current is increased, the lines of force move outwards from the coil. If the current is reduced, the lines of force move inwards. If another coil is placed adjacent to the first coil then, as the field moves out or in, the moving lines of force will "cut" the turns of the second coil. As it does this, a voltage is induced in the second coil. With the 50 Hz AC mains supply, this will happen 50 times a second. This is called MUTUAL INDUCTION and forms the basis of the transformer.

Rectifier:

A rectifier is an electrical device that converts alternating current (AC) to direct current (DC), a process known as rectification. Rectifiers have many uses including as components of power supplies and as detectors of radio signals. Rectifiers may be made of solid-state diodes, vacuum tube diodes, mercury arc valves, and other components. A device that it can perform the opposite function (converting DC to AC) is known as an inverter. When only one diode is used to rectify AC (by blocking the negative or positive portion of the waveform), the difference between the term diode and the term rectifier is merely one of usage, i.e., the term rectifier describes a diode that is being used to convert AC to DC. Almost all rectifiers comprise a number of diodes in a specific arrangement for more efficiently converting AC to DC than is possible with only one diode. Before the development of silicon semiconductor rectifiers, vacuum tube diodes and copper (I) oxide or selenium rectifier stacks were used.

Filter:

The process of converting a pulsating direct current to a pure direct current using filters is called as filtration. Electronic filters are electronic circuits, which perform signal-processing functions, specifically to remove unwanted frequency components from the signal, to enhance wanted ones.

Regulator:

A voltage regulator (also called a ‗regulator') with only three terminals appears to be a simple device, but it is in fact a very complex integrated circuit. It converts a varying input voltage into a constant ‗regulated 'output voltage. Voltage Regulators are available in a variety of outputs like 5V, 6V, 9V, 12V and 15V. The LM78XX series of voltage regulators are designed for positive input. For applications requiring negative input, the LM79XX series is used. Using a pair of ‗voltage-divider' resistors can increase the output voltage of a regulator circuit. It is not possible to obtain a voltage lower than the stated rating. You cannot use a 12V regulator to make a 5V power supply. Voltage regulators are very robust. These can withstand over-current draw due to short circuits and also over-heating. In both cases, the regulator will cut off before any damage occurs. The only way to destroy a regulator is to apply reverse voltage to its input. Reverse polarity destroys the regulator almost instantly. Fig: 3u shows voltage regulator.

Spirometer:

A **spirometer** is a medical device used to measure **lung function** by assessing the volume of air a person can inhale and exhale. It is primarily used to diagnose and monitor **respiratory diseases** such as **asthma**, **chronic obstructive pulmonary disease (COPD)**, and other pulmonary conditions. Spirometry is a critical diagnostic tool in **pulmonology** and is often part of routine checks for people with chronic respiratory diseases or those at risk.

Software Requirements

Arduino IDE:

A program for Arduino may be written in any programming language for a compiler that produces binary machine code for the target processor. Atmel provides a development environment for their microcontrollers, AVR Studio and the newer Atmel Studio.

The Arduino project provides the Arduino integrated development environment (IDE), which is a cross-platform application written in the programming language Java. It originated from the IDE for the languages Processing and Wiring. It includes a code editor with features such as text cutting and pasting, searching and replacing text, automatic indenting, brace matching, and syntax highlighting, and provides simple one-click mechanisms to compile and upload programs to an Arduino board. It also contains a message area, a text console, a toolbar with buttons for common functions and a hierarchy of operation menus.

A program written with the IDE for Arduino is called a sketch. Sketches are saved on the development computer as text files with the file extension .ino. Arduino Software (IDE) pre-1.0 saved sketches with the extension .pde.

The Arduino IDE supports the languages C and C++ using special rules of code structuring. The Arduino IDE supplies a software library from the Wiring project, which provides many common input and output procedures. User-written code only requires two basic functions, for starting the sketch and the main program loop, that are compiled and linked with a program stub main() into an executable cyclic executive program with the GNU toolchain, also included with the IDE distribution.
A minimal Arduino C/C++ sketch, as seen by the Arduino IDE programmer, consist of only two functions:

- **setup():** This function is called once when a sketch starts after power-up or reset. It is used to initialize variables, input and output pin modes, and other libraries needed in the sketch.
- **loop():** After setup() has been called, function loop() is executed repeatedly in the main program. It controls the board until the board is powered off or is reset.

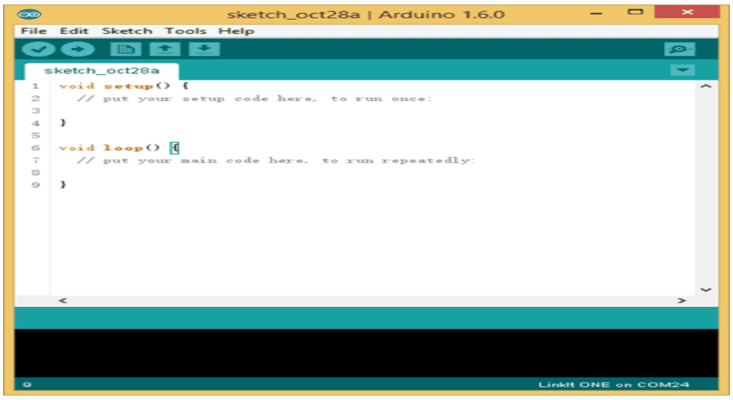


Fig . An Arduino Sketch

Embedded C:

- When designing software for a smaller embedded system with the 8051, it is very common place to develop the entire product using assembly code. With many projects, this is a feasible approach since the amount of code that must be generated is typically less than 8 kilobytes and is relatively simple in nature. If a hardware engineer is tasked with designing both the hardware and the software, he or she will frequently be tempted to write the software in assembly language.
- The trouble with projects done with assembly code can is that they can be difficult to read and maintain, especially if they are not well commented. Additionally, the amount of code reusable from a typical assembly

language project is usually very low. Use of a higher-level language like C can directly address these issues. A program written in C is easier to read than an assembly program.

- Since a C program possesses greater structure, it is easier to understand and maintain. Because of its modularity, a C program can better lend itself to reuse of code from project to project. The division of code into functions will force better structure of the software and lead to functions that can be taken from one project and used in another, thus reducing overall development time. A high order language such as C allows a developer to write code, which resembles a human's thought process more closely than does the equivalent assembly code. [25]The developer can focus more time on designing the algorithms of the system rather than having to concentrate on their individual implementation. This will greatly reduce development time and lower debugging time since the code is more understandable.
- By using a language like C, the programmer does not have to be intimately familiar with the architecture of the processor. This means that someone new to a given processor can get a project up and running quicker, since the internals and organization of the target processor do not have to be learned. Additionally, code developed in C will be more portable to other systems than code developed in assembly. Many target processors have C compilers available, which support ANSI C.
- All of this is not to say that assembly language does not have its place. In fact, many embedded systems (particularly real time systems) have a combination of C and assembly code. For time critical operations, assembly code is frequently the only way to go. One of the great things about the C language is that it allows you to perform low-level manipulations of the hardware if need be, yet provides you the functionality and abstraction of a higher order language.

CHAPTER 5

APPLICATIONS

1. **Chronic respiratory disease monitoring:** Used for continuous monitoring of conditions like Asthma, COPD (Chronic Obstructive Pulmonary Disease), and Pulmonary Fibrosis. Helps track lung function over time and adjust treatment plans accordingly.
2. **Home-Based Monitoring:** Enables patients to monitor their lung health at home, reducing hospital visits. Provides real-time data that can be shared with healthcare providers remotely.
3. **Telemedicine:** Facilitates remote diagnosis and management of respiratory conditions through data sharing with doctors or clinics. Allows healthcare providers to monitor patient progress and intervene if necessary.
4. **Sports and Fitness:** Used by athletes to assess their lung capacity and optimize training regimens. Helps in improving performance by monitoring breathing patterns and recovery times.
5. **Workplace Health Monitoring:** Used in industries where workers are exposed to hazardous environments (e.g., mining, construction) to track lung health. Prevents long-term respiratory issues by offering early detection of abnormality.

CHAPTER 6

8.1 CONCLUSION AND FUTURE SCOPE:

The Smart Spirometer designed using an STM32 microcontroller, BMP280 pressure sensor, and IR sensors provides an efficient, low-cost, and portable solution for measuring lung capacity and assessing respiratory health. The system successfully integrates multiple sensors to measure exhalation pressure and airflow, which are key indicators of lung function. By displaying real-time results on an LCD, the device empowers users to monitor their lung health conveniently at home or in a clinical setting. The main advantages of this system include: Accuracy: The combination of the BMP280 sensor for pressure measurement and IR sensors for lung capacity detection ensures precise results, making it a reliable tool for evaluating lung function. Affordability: By using commercially available sensors and a low-cost microcontroller, the spirometer can be produced at an affordable cost, making it accessible for personal and healthcare applications. Portability: The compact design of the spirometer allows users to carry it for on-the-go monitoring, offering flexibility for patients and healthcare providers alike. Real-time Monitoring: The system provides immediate feedback to the user, enabling quick assessments of lung health, which is crucial for early detection of respiratory issues. Data Logging and Alerts: The system can store data for future reference and can send alerts if abnormal lung capacities are detected, improving ongoing health management. Future improvements could include the addition of wireless connectivity (e.g., Bluetooth or Wi-Fi) for remote monitoring, integration with healthcare apps, and further sensor calibration for more advanced respiratory diagnostics. Overall, the Smart Spirometer offers a promising, accessible solution for proactive respiratory health monitoring, helping individuals better understand and manage their lung health while facilitating easier and more affordable access to spirometry.

8.2 Future scope:

1. Integration with AI and Machine Learning: The future of smart spirometers lies in advanced AI algorithms that can analyze lung function data in real-time to detect early signs of respiratory conditions, predict disease progression, and offer personalized treatment recommendations.

2. Improved Accuracy and Sensitivity: Future models will see improvements in sensor accuracy, enabling more precise measurements and better differentiation between different types of lung disorders.

3. Wearable Integration: Future smart spirometers could be integrated into wearable devices, allowing for continuous, real-time lung function monitoring without the need for periodic tests.

4. Miniaturization and Portability: The development of smaller, more compact spirometers with longer battery life will further enhance the portability and accessibility of the device, especially for use in remote areas and during travel.

5. Telemedicine and Real-Time Doctor Feedback: Integration with telemedicine platforms will enable direct communication between patients and healthcare professionals, with immediate feedback and personalized care recommendations based on real-time data.

6. Global Health Applications: With the ability to monitor lung health remotely, smart spirometers can be pivotal in improving respiratory care in regions with limited access to healthcare, enabling widespread early detection and prevention of respiratory diseases.

CHAPTER 7

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CHAPTER 8

RESULT ANALYSIS